Visual grounding promotes verb learning in low data conditions. Introducing V-CLIP

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Abstract

Visual grounding yields advantages in the acquisition of verb meanings when linguistic data are scarce. We investigate this claim by training a CLIP-style dual-encoder model on the MSVD video-caption corpus under low-data constraints, comparing (i) a video-grounded model (V-CLIP) that samples eight frames per video, (ii) a standard CLIP model using only one single central frame. All models are trained on progressively larger subsets of MSVD captions (ranging from 15K to 75K tokens), with hyperparameters held constant across conditions. We evaluate verb representations on SimVerb-3500, and assess general word encoding using WordSim353. Results show that V-CLIP outperforms CLIP in correlating with human verb similarity, and also yields gains on WordSim353. Ablation of frame count reveals that eight sampled frames provide a balance between dynamic framing and noise. Our results constitute early evidence that dynamic grounding enhances verb embedding quality in data-limited regimes, suggesting a more cognitively plausible route to efficient verb learning.

1 Introduction

Large language models (LLMs) consume orders of magnitude more text than a human child ever hears: typically tens of billions of sentences, whereas children receive roughly a few dozens of million of sentences in their first three years of life [1][2]. This stark disparity raises doubts about the LLMs' usefulness as models of human language acquisition. Instead, one plausible path toward human-like language learning is to ground words in the perceptual stream available to a learner [3][4]. Very recent work shows that in low-data regimes, pairing static images with captions accelerates noun acquisition, but offers little benefit for other parts of speech [5][6]. Verbs, whose semantic content often revolves around temporally extended events, may demand non-static visual evidence. We therefore ask: Does training a CLIP-style vision–language model on video–caption pairs lead to verb embeddings that align more closely with human similarity judgments than the same model trained on single frames from those videos? The objective is twofold: data efficiency as well as cognitive and developmental plausibility, which is achieved by mirroring the bimodal experience of early learners. By isolating verbs and strictly controlling training data size, our study intends to explore a new method that adds up to a growing new line of research.

English verbs vary widely in meaning, but a large portion of their semantics is devoted to actions and events. The long-standing Princeton WordNet lexical database [7] groups verbs into broad semantic categories and most involve actions or events: *verb.motion*, *verb.contact verb.creation verb.consumption verb.communication verb.social*. Neuroscience has even revealed an embodied basis for action-verb semantics. A famous and seminal fMRI experiment [8] found that reading action-related words activates motor-planning and sensorimotor brain regions in patterns reflecting the described action. For example, reading verbs such as *kick*, *pick*, or *lick* triggers somatotopic activity in primary motor and premotor cortices; notably, leg-related regions for "kick" versus tongue-related regions for "lick" [9]. Moreover, in itself action recognition appears to be a domain-specific capacity for which humans have evolved specialized resources: MEG studies demonstrate that the brain integrates spatiotemporal features to decode actions invariantly within approximately 200 ms of video onset, and that both form and motion cues are required for this rapid, invariant recognition [10].

2 Background

Children master core lexical meanings with astonishing speed: by six months, many common nouns are already understood [11], and productive vocabularies begin to emerge around the first year mark. Psycholinguistic work suggests that this efficiency stems in large part from grounding words in concurrent perceptual experience [12]. Because infants learn many verbs from sparse contexts, verb acquisition under severe data limits provides a stringent test-bed for computational models of lexicon development.

[1]Contributions. The authors collaborated closely on the project; however, Ferdinando Pio D'Elia mainly focused on conceptualization, literature review, and manuscript writing, while Tjaard Pfitzner mainly focused on code development, model training, and conceptualization.

Recent multi-modal pre-training methods supply large-scale visual grounding that could narrow the data efficiency gap. Among them, CLIP aligns 300M image–caption pairs via a contrastive objective and yields transferable word representations [13][14]. Other fusion strategies include the generative GIT model, which conditions next-token prediction on CNN features, and Flamingo, which modulates a frozen language backbone with learned cross-modal attention. However, evaluations in low-data regimes show only conditional benefits [15], and no study has isolated verb representations. Our work therefore asks whether explicitly grounding verbs through the CLIP objective accelerates word-meaning acquisition when training data are scarce.

It is also worth noting that complementary evidence from neuroscience points to developmental parallels between children and transformer-based language models. Evanson, Lakretz & King [16] trained 48 GPT-2 instances from scratch and found that, like children aged 18 months-6 years, models acquire linguistic skills in a systematic order: phoneme discrimination improves earliest, lexical tasks later, and complex syntax last; yet learning is "parallel" in that later stages begin to improve from the first updates, revealing both alignments and divergences between biological and artificial trajectories. Extending this behavioural link to the brain, a recent intracranial preprint [17] recorded more than 7,400 electrodes in 46 participants aged 2–46 years listening to linguistic input. They showed that fast phonetic codes are robust in the superior temporal gyrus even in toddlers, whereas slower lexical representations emerge only in association cortices of older children; strikingly, the same hierarchy unfolds during LM training, with deep wav2vec 2.0 and Llama-3 layers aligning exclusively to adult-like cortical responses. These recent findings seem to strengthen the case for grounded neural networks as mechanistic models of human language acquisition.

3 Methods

We here describe the models, baseline, and dataset used to evaluate visual grounding under low-data conditions. Our approach extends CLIP to video (V-CLIP) and compares it against a GloVe baseline using the MSVD corpus.

3.1 Models

Our implementation follows the CLIP paradigm [13], employing a dual-encoder architecture that jointly learns visual and textual representations in a shared 512-dimensional space. Depending on whether we run in "CLIP" (image) mode or "V-CLIP" (video) mode, the visual branch processes either a single central frame or multiple uniformly sampled frames from each video, respectively. Both variants share the same text branch, backbone, and contrastive-loss objective; only the image preprocessing differs. Both architectures are also outlined in Figure 1 and Figure 2.

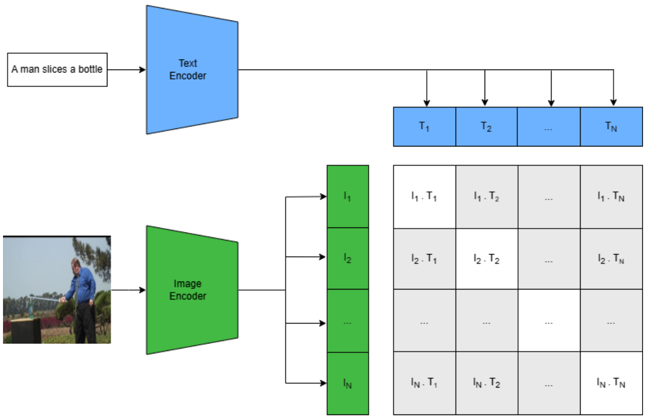

Figure 1: CLIP architecture: a dual-encoder model jointly learns image and text representations by processing a single video frame through a ResNet-18 backbone and projecting it into a shared 512-dimensional space.

3.1.1 Image encoder

In our single-frame CLIP model, we extract one central frame from each video, resize it to 224×224 pixels and feed it into a ResNet-18 backbone (classification head removed), yielding a (512-D if including projection head) feature vector after global pooling and projection. A lightweight projection head (dropout → linear layer) then maps the pooled ResNet output to a 512-D embedding. This single-frame embedding therefore represents the entire input in CLIP.

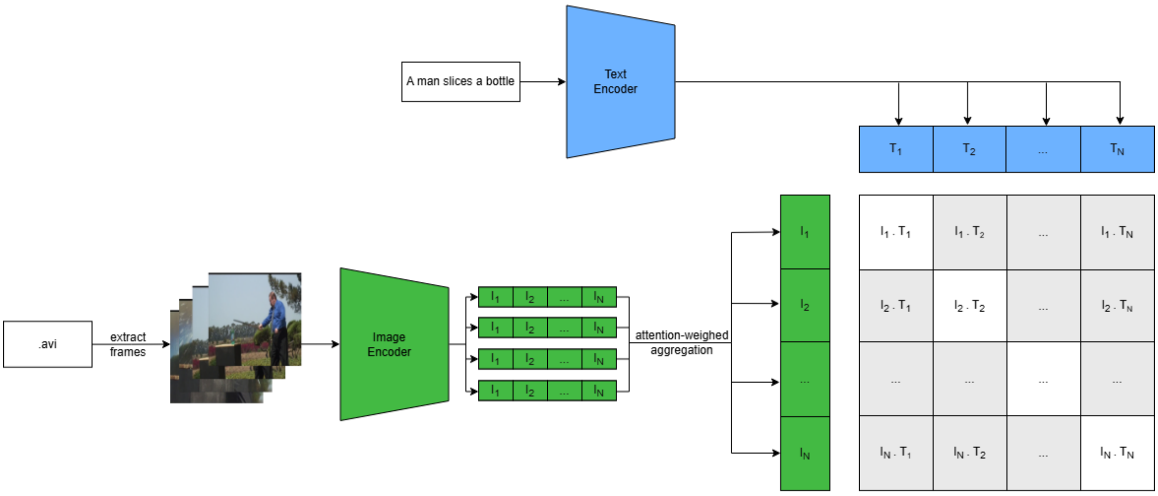

Figure 2: V-CLIP architecture: extending CLIP to video by encoding multiple uniformly sampled frames through a ResNet-18 backbone and aggregating them via attention, yielding temporally grounded 512-dimensional embeddings aligned with text.

In our multiframe V-CLIP model, inspired by the intuition that in videos, the informative peak of an action tends to occur after its very beginning and before its very end, we first restrict frame sampling to the "inner" 60% (ie. we prune the first and last 20%). From this remaining segment, we uniformly sample T RGB frames (in our experiments, $T = 8$ unless otherwise noted) from each video, resize to 224×224, and pass each through a ResNet-18 backbone (classification head is removed). Moreover, per-frame features (512-D after global average pooling) are temporally aggregated via attention-weighted pooling over those T frames, returning one 512-D vector for the whole clip. A projection head (dropout $\rightarrow$ linear layer) maps that aggregated vector to a 512-D learnable embedding, which we ℓ_2-normalize before computing the contrastive-loss.

3.1.2 Text encoder

In our text branch, we begin by lowercasing each caption and running it through DistilBERT's tokenizer, padding or truncating every sequence to exactly 32 tokens so that we produce both token IDs and attention masks. Those IDs and masks go straight into a pre-trained DistilBERT model, from which we pull out the pooled [CLS] representation (a 768-dimensional vector). We then apply a lightweight projection head (first a dropout layer and then a linear layer) to convert that 768-dimensional [CLS] vector into a 512-dimensional embedding. Finally, before any contrastive comparisons happen, we ℓ_2-normalize that 512-dimensional text embedding so that it is compatible with the visual embeddings when computing cosine similarities for the loss.

3.1.3 Contrastive training

Within each minibatch, we compute cosine similarities between all visual (image or video) embeddings and all textual embeddings, scaling by a fixed temperature ($\tau = 0.07$). There, we apply a symmetric cross-entropy loss: one term treats visual$\rightarrow$text as positive, the other text$\rightarrow$visual. In practice, each matching pair should have high similarity, while non-matching pairs are pushed apart. At training/inference time, we switch pre-processing based on the model we intend to use and proceed as outlined above. The following settings remain identical across the two models: DistilBERT text encoder, projection heads, ℓ_2-normalization, and contrastive-loss hyperparameters.

3.2 GloVe baseline

As a non-grounded point of comparison, we implement a classic Global Vectors for Word Representation (GloVe) [18] model using our caption corpus. We include GloVe only as a trivial, static-embedding baseline, since our very aim is to compare CLIP against the newly setup V-CLIP. Furthermore, prior work has already shown that, under low data conditions, a CLIP-style model trained with static images surpasses language-only models [5]: such prior research was carried out on GPT-2–based contextual embeddings, the effectiveness of which we therefore take as established for the purposes of the present study. Against that backdrop, we keep GloVe only as a sanity-check baseline, illustrating how performance collapses once contextual information is stripped away. GloVe's poor performance is unsurprising given that: (i) each word is represented by a single, context-agnostic vector that is unable to capture verb polysemy; (ii) our caption corpus (41K sentences) is too small to produce reliable co-occurrence statistics for many words, leading to extremely noisy embeddings. Moreover, preliminary experiments indicated that training GloVe with one, three, five, or all available captions per video in the MSVD dataset did not yield any significant impact on its performance scores. To set up the model, we first build a vocabulary by retaining every token that appears at least once in the training captions. Next, we construct a sparse co-occurrence matrix by sliding a context window of size 5 over each tokenized sentence, weighting each word–context pair's count by the inverse of their distance within that window. The model comprises two embedding matrices—one for "target" words and one for "context" words—each of dimension d, along with bias vectors. During optimization, the dot product of a target–context embedding pair (plus their biases) is trained to match the logarithm of the corresponding co-occurrence count, using the GloVe weighting function $\left(\min\{x/x_{\max}, 1\}\right)^{\alpha}$ (with $x_{\max} = 100$ and $\alpha = 0.75$) to down-weight very frequent co-occurrences. At the end of training, we sum each word's target and context embeddings to obtain its final representation. We initialize both embedding matrices via Xavier uniform and set all bias terms to zero. We optimize the weighted least-squares objective using the Adam optimizer with a learning rate of 0.005. Training proceeds for 50 epochs over the entire co-occurrence matrix. After optimization, we extract each word's final embedding by summing its target and context vectors; these embeddings are then used for all downstream evaluations.

3.3 Dataset

The MSVD Video Caption (MSVD; [19]) contains 1,970 open-domain YouTube clips of approximately 3–10s each, accompanied by about 41K crowdsourced English captions. We partition the videos into an 80% train / 20% validation split, ensuring that all captions associated with a given video remain within the same partition. For each training instance, the dataloader first selects one caption at random from those available for the video. It then decodes the video and uniformly samples T=8 RGB frames (reduced to T=1 in the single-frame control), resizes every frame to 224×224, converts them to tensors, and caches the result for subsequent epochs. Finally, the chosen caption is lower-cased and tokenized with DistilBERT, with sequences padded or truncated to 32 tokens so that token-ID and attention-mask tensors are produced. The loader therefore outputs a tuple (F×C×H×W tensor, token IDs, attention mask) that feeds directly into CLIP training. The identical caption corpus also drives the GloVe baseline to match linguistic input across grounded and non-grounded methodologies. Examples of captions from the dataset are given in Appendix B. While occasional grammatical, lexical, or spelling errors can be found in the captions for a given video, such cases are relatively rare. Moreover, because these imperfections are uniformly distributed across all three models (V-CLIP, CLIP, and GloVe), we argue that they have a negligible effect and do not compromise the validity of our findings.

4 Experimental set-up

In this section we outline the training protocols and parameters (Section 4.1 and then proceed to provide an account of the evaluation benchmarks we use to assess model performance (Section 4.2.

4.1 Training details

We train the CLIP and V-CLIP models for multiple epochs with different hyperparameter configurations. Changes are the number of captions per image/video, the number of frames used in V-CLIP or the temperature parameter that controls the sharpness of the similarity distribution between pairs of embeddings. For the baseline GloVe model, we also train for multiple epochs, changing the hyperparameter configuration; however, as we saw no significant changes in the performance on the benchmarks, we will not include explorative results in this paper. The code for training and evaluation can be found at: https://github.com/Tjaardo/CLIP-V-CLIP.

4.2 Evaluation

In this section, we give an account of the benchmarks we adopt, aligning with ample literature in the field. We evaluate all models using two human-rated word-similarity datasets: SimVerb-3500 and WordSim353. First, we extract each word's embedding from a trained model and compute cosine similarity for every pair in the benchmark. Then, we measure how well these cosine similarities align with the human-annotated scores by calculating Spearman's rank correlation. SimVerb-3500 [20] contains 3500 verb pairs, each rated for semantic similarity by human annotators; examples are shown in Appendix C. Unlike general-purpose similarity datasets, SimVerb-3500 focuses exclusively on verbs and covers all normed verb types from the USF free association database, the largest free association database in English [21]. We selected SimVerb-3500 because most similarity benchmarks emphasize nouns and this results in verbs being underrepresented; instead, this dataset aims to fill that gap on verb semantics. On the other hand, WordSim353 [22] consists of 353 word pairs, including a widespread mix of nouns, verbs, and adjectives, covering both concrete and abstract concepts. Each word pair is rated by human annotators based on how semantically similar the two words are. This benchmark is used to evaluate the word learning capabilities of the models, not only on verbs but on a general vocabulary that does not differentiate between lexical categories.

5 Results and discussion

In this section, we present the results of the three models - GloVe, CLIP, and V-CLIP - on the two similarity benchmarks: SimVerb-3500 and WordSim353. To improve the performance of the V-CLIP model, which is the main focus of this paper, we adjust the hyperparameter configuration and then evaluate the results on our benchmarks. All models were trained for multiple epochs, and evaluation results were averaged across multiple runs to mitigate the impact of outliers.

5.1 In verb semantics, V-CLIP outperforms CLIP

Figure 3 shows the evaluation of the different models on both benchmarks. Our results suggest that utilizing visual grounding via CLIP greatly enhances semantic similarity predictions compared to word-only models like GloVe, which is vastly expeceted. Additionally, incorporating temporal visual grounding with V-CLIP yields additional improvements on not only verb similarity but also general word similarity benchmarks. The gains are limited but significant: V-Clip consistently outperforms CLIP by ≈0.05 during verb-specific evaluation; V-Clip moderately outperforms CLIP by ≈0.02 during verb-specific evaluation on the broader WordSim353 benchmark.

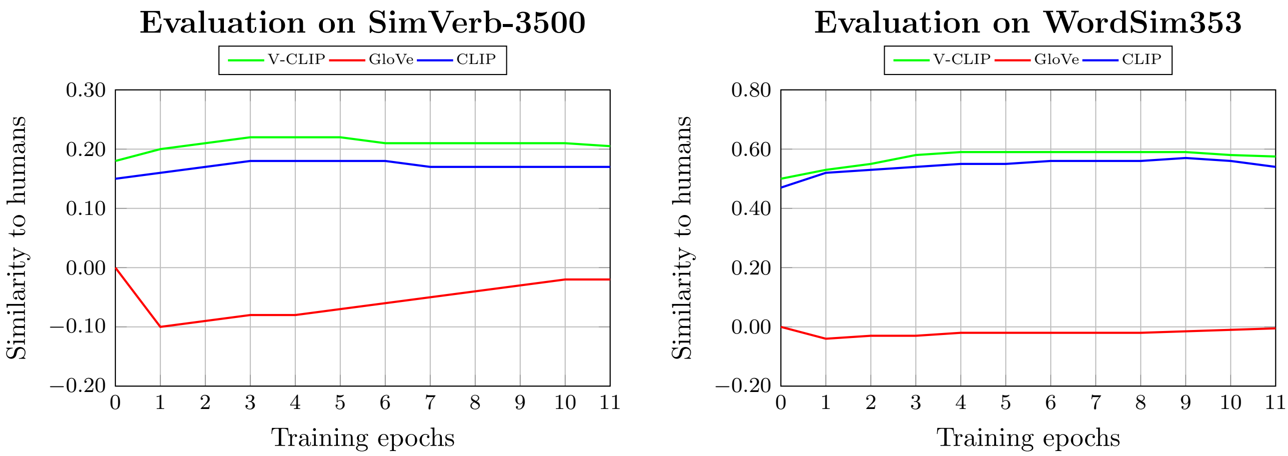

Figure 3: Average word learning performance of different models on word similarity benchmarks. (left) Average Spearman correlation between the cosine similarity of word embeddings from models and the SimVerb-3500 benchmark. CLIP and V-CLIP greatly outperform the word-only GloVe model, with V-CLIP showing the highest similarity to humans. (right) Average Spearman correlation between the cosine similarity of word embeddings from models and the WordSim353 benchmark. We find similar results to those shown in (left). GloVe is not able to learn word meanings, while CLIP and V-CLIP achieve a fairly high average similarity to humans, with V-CLIP outperforming CLIP.

Despite the improvements brought by visual grounding in CLIP and V-CLIP, the overall performance on the benchmarks remains relatively low. We attribute this to two primary factors. First, the models are trained under low-data conditions, limiting their ability to generalize semantic relationships effectively. Second, there are inherent limitations in the benchmarks themselves. SimVerb-3500 contains a wide range of verb pairs, many of which likely include verbs that the models have not encountered during training, making it challenging to achieve high similarity correlations. In contrast, WordSim353 has a much smaller set of 353 word pairs with a general mix of parts of speech. While its limited size reduces vocabulary diversity, it also increases the likelihood that the models have seen the benchmark words during training, which explains the comparatively higher scores we observed in our experiments. Our results align with earlier work [5][6]: CLIP's performance gains are most pronounced when the evaluation targets nouns. This category-specific boost we found, reinforces the broader claim that lexical-contrastive grounding does not affect all word classes equally; the impact depends on the kind of grounding signal supplied (e.g., static or dynamic visual input).

5.2 Performance peaks with an input of 8 frames

To gain deeper insights into the relative contributions of the static and temporal visual grounding, we alter the number of frames used in the training process. If for a single video not enough frames are available (after the 60% pruning we described above), we duplicate them to get to the required number. Figure 4 visualizes the performance impact of frame numbers on the two similarity benchmarks. We find that using eight video frames results in the highest correlation with human judgments, surpassing both smaller (2-4 frames) and larger (16 frames) configurations. These experiments therefore found an *optimum* point where temporal information is sufficient and excessive noise is prevented.

In our experiments, however, we do not attempt a frame selection that is driven by representativeness; instead, for each video-caption pair, we pick the middle frames (i.e., evenly spaced frames centered around each clip) through attention-weighted aggregation. This decision is intentional: we adopt this straightforward sampling strategy to test whether dynamic context alone can already boost performance without finely selecting which frames might be most informative. Our results therefore seem to indicate that even unoptimized video input provides enough dynamic information to improve verb embedding quality under low-data conditions.

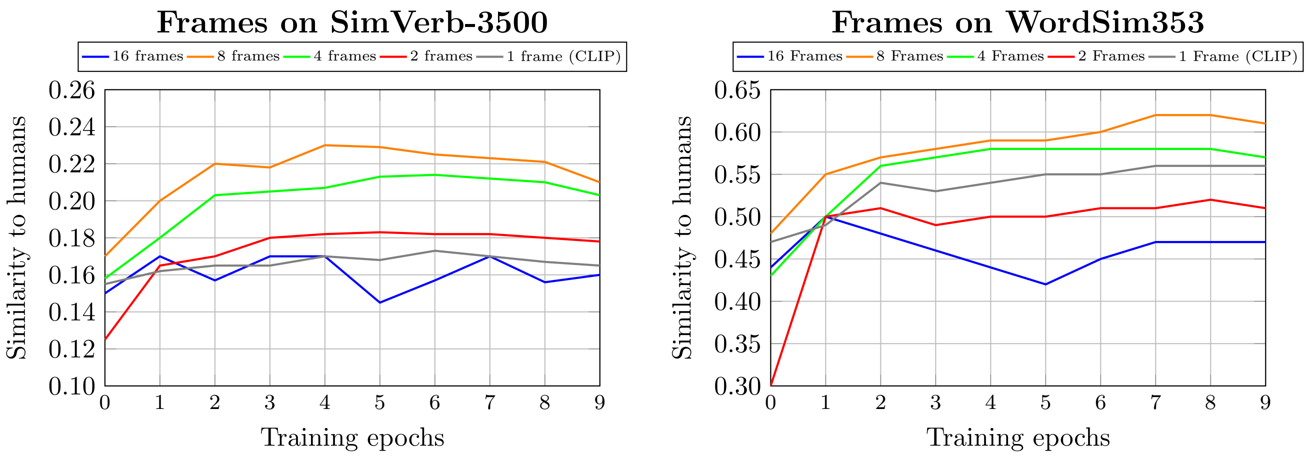

Figure 4: Average evaluation results on the SimVerb-3500 benchmark (left) and the WordSim353 benchmark (right) across training epochs for different numbers of input frames. Using more frames generally results in a better performance on both benchmarks, with 8-frame and 4-frame configurations achieving the highest similarity to human judgments. However, increasing the frame number further to 16 leads to a significant loss in performance, indicating introduced noise or redundancy.

5.3 Temperature of 0.7 is confirmed ideal

We also explore the impact of the temperature parameter during contrastive learning with V-CLIP. In the original CLIP's contrastive-learning framework, the temperature τ acts as a scaling factor on the cosine-similarity logits before the softmax. Concretely, lowering τ "sharpens" the similarity distribution: high-scoring image–text pairs receive even larger softmax probabilities, making the model more selective, whereas raising τ smooths the distribution, allowing more pairs to share probability mass. As seen in Figure 5, decreasing the temperature parameter consistently improves performance on both the SimVerb-3500 and the WordSim353 benchmarks. The best results are achieved with a temperature of 0.07, suggesting that more selective alignment between visual and textual modalities benefits more human-like semantic representations. Our results align with the original CLIP paper [13], as a temperature value of 0.07 is recommended as it allows a good balance between training stability and semantic clarity of embeddings [23].

5.4 Increasing the number of captions/tokens hinders performance

To assess the impact of semantic diversity on the training data, we vary the number of distinct captions used per video. While the visual input remains the same, each video is paired with one, three, or five different captions throughout training. Results show that using only one caption per video leads to the best result on both the SimVerb-3500 and the WordSim353 benchmarks (see Figure 6). Adding more captions tends to negatively affect performance as training progresses. We hypothesize that this drop is due to the high similarity among the captions. Instead of introducing genuinely new semantic information, the additional captions may encourage the model to overfit to superficial linguistic patterns, ultimately reducing its ability to generalize. We show the increased overfitting behavior with more captions in Figure 7.

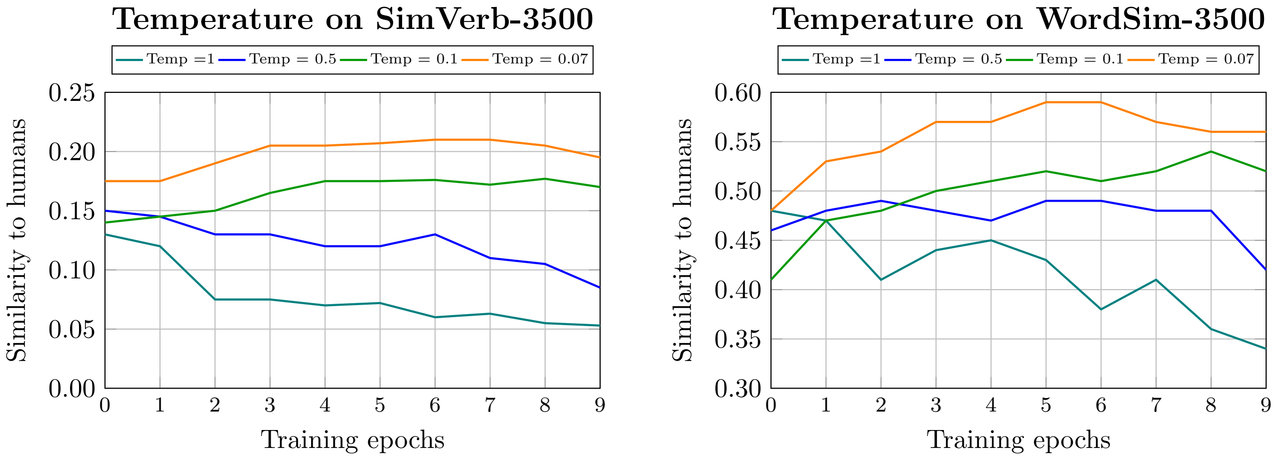

Figure 5: Average evaluation results on the SimVerb-3500 benchmark (left) and the WordSim353 benchmark (right) across several temperature settings. Setting temperature to 0.7 results in better performance on both benchmarks, aligning with earlier studies [23][13].

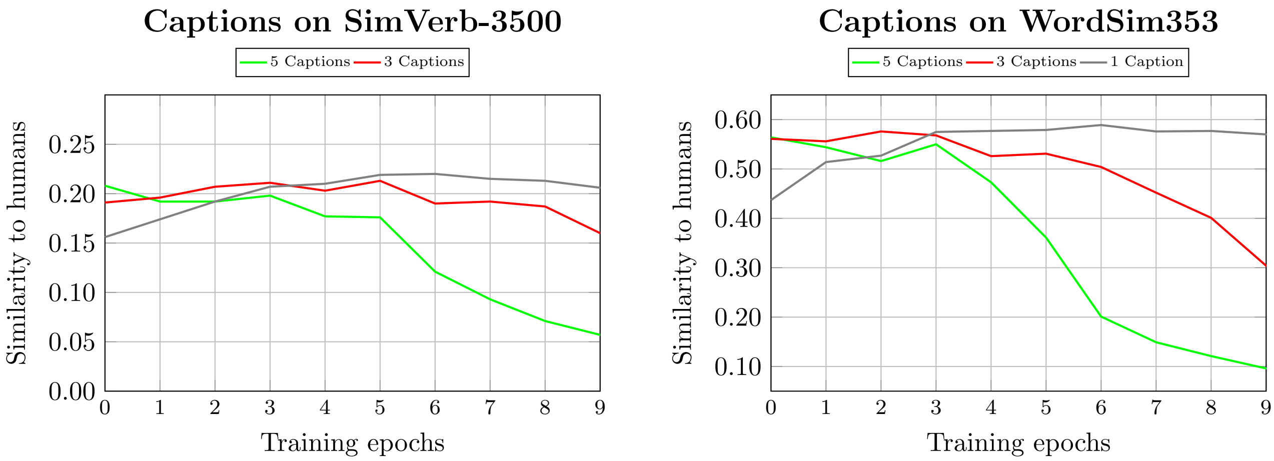

Figure 6: Effects of the number of unique captions per video on performance across training epochs. Models trained on only one caption per video consistently outperform those trained with three or five, across both benchmarks. Increasing the number of captions appears to encourage overfitting and reduce generalization, resulting in decreased performance for longer epochs.

Clearly, the 3-caption (red) and 5-caption (green) curves both bottom out around epoch 2 and then steadily increase in validation loss, whereas the 1-caption (gray) curve continues to decline. This divergence coincides exactly with the drop in correlation seen in Figure 6. In other words, extra (and highly similar) captions cause the model to start memorizing redundant linguistic patterns rather than learning broad verb semantics. As a result, models trained with three or five captions per video overfit much earlier, which explains their poorer performance on both SimVerb-3500 and WordSim353.

6 Limitations

The work we present in this report is subject to certain limitations, primarily due to resource and time constraints. Notably, our experimental scope is restricted to a single family of models, CLIP, while excluding other promising architectures such as GIT, Flamingo, and similar vision–language models. Additionally, the evaluation is limited to a specific type of benchmark, leaving broader evaluation opportunities unexplored. In particular, both SimVerb-3500 and WordSim353 are non-contextual lexical similarity datasets; employing contextualized benchmarks (e.g., sentence-level or usage-based evaluations) could more thoroughly test how well the embeddings capture word meaning in context. Constrained by time, we also limit training to around 10 epochs for each model; a longer training process might yield better convergence and more robust results. Furthermore, the type of videos and visual data we use in our experiments are not necessarily representative of what children are typically exposed to during natural language learning; not to mention an even deeper limitation: humans perceive and integrate information from multiple modalities beyond vision and hearing,

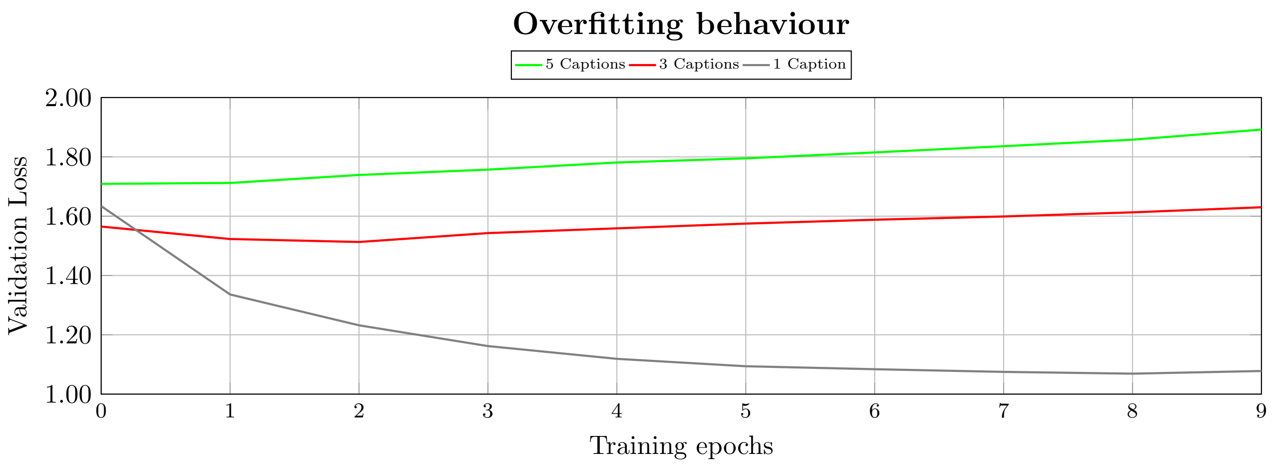

Figure 7: Validation loss of the V-CLIP model with different numbers of captions per video across training epochs. Models trained with three or five captions per video show a significant increase in validation loss over time, indicating overfitting likely caused by redundant or highly similar captions.

such as smell, taste, and touch. As such, decades of work are still needed to validate that these algorithms can learn high-quality word representations from the same distribution of sensory and visual inputs that children receive.

Beyond these broader concerns, our dataset itself is very small: although MSVD provides roughly 1,970 videos and around 41K captions, our low-data regime experiments often used only a fraction of that vocabulary, limiting generalizability. Relatedly, the captions themselves tend not to be very diverse: in many cases, multiple captions for the same video exhibit high lexical overlap; the model may therefore overfit to recurrent phrases or fail to encounter sufficient linguistic variety to learn robust representations. We acknowledge all these limitations but consider them as natural directions for future research.

7 Future work

Building on our findings, we see several concrete avenues to address the limitations above as well as to extend this line of research:

(a) Motivated selection of data. Rather than sampling frames at random or always using the central frames, future work could implement a supervised selection method. An option might be to train a saliency model to pick the most informative T frames per video, rather than relying on uniform sampling. Additionally, low-quality or near-duplicate frames might be pruned before presenting them to the backbone. Another option might involve filtering the captions via fluency or grammar-based classification. As discussed, this could help mitigate overfitting on repetitive or noisy textual data.

(b) Advanced attention pooling. Although our current V-CLIP aggregates per-frame ResNet-18 features via a simple attention-weighted pooling, future work could explore more sophisticated attention mechanisms (e.g., multi-head temporal attention or transformer-based fusion) to dynamically weight frames based on their semantic relevance to the accompanying caption. A promising idea could be to fine-tune a lightweight transformer that attends across the T per-frame embeddings before passing a single fused vector to the projection head. This may significantly reduce noise from uninformative frames.

Our results suggest that dynamic visual grounding benefits certain parts of speech (ie. verbs) more than others. We believe future work should investigate what factors might drive these category-specific differences: is the improvement driven purely by semantics (i.e., verbs often denote actions, so temporal information is crucial), or are there other syntactic and pragmatic factors at play? Future directions might also explore how perceptual affordances (e.g., the embodied nature of actions) shape the alignment between vision and language, and how the addition of auditory streams (e.g., speech audio, ambient sounds) might contribute to word learning. We believe that a systematic investigation might not also improve training efficiency in language models, but - more importantly - also gain a better insight into the cognitive principles that underlie grounded word learning. Such an investigation should be developed at the intersection of developmental psycholinguistics, formal linguistics, and multimodal representation learning.

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A Samples of video-caption pairs from dataset

(a) 1 Frame

(b) 2 Frames

(c) 4 Frames

(d) 8 Frames

(e) 16 Frames

Figure 8: Extracted frames from an example video during the training process with different frame-number configurations.

B Samples of captions from the MSVD dataset

Table 1: We present a sample of captions from the MSVD dataset, highlighting in red the captions that generate grammatical and/or lexical error.

Item label	Captions
4wsuPCjDBc_5_15	a squirrel is eating a peanut in its shell / a chipmunk is eating / a chipmunk is eating a peanut / a chipmunk is eating a nut / a squirrel is eating a nut / a squirrel is eating a whole peanut / a squirrel is eating a peanut / a squirrel is eating nuts / a small squirrel is eating a peanut / a small animal is chewing on a nut / an ardilla is eating / a squirrel is eating a nut / a chipmunk is eating some food / the squirrel is eating / a ferret eats a leaf / a hamster is eating a peanut / a ardilla coreana eating / the rabbit is eating / a chipmunk is eating a peanut / a rabbit is eating / a rabbit is eating / a squirrel eating a fruit / a chipmunk is eating / a man is playing guitar / a squirrel or some such animal eating a nut of some kind / a chipmunk is eating some food / a rat is playing and eating something / the squirrel ate the peanut out of the shell / the squirrel is eating
Cv5LsqKUXc_17_25	a man is slicing strawberry / a men is removing a stem of strawberry for making ichigo daifuku strawberry daifuku / a woman is removing stems from a strawberry / a woman is slicing a strawberry / chopping stroubery / someone is cutting the leafy parts off of some strawberries / the stem / she teaching strobery food making / a chef cuts the leaves off of strawberries / a chef is cutting strawberries head / a cook is cleaning strawberry by removing the leaves / a man cuts the leaves off of strawberries / a man cuts the tops off of three strawberries / a man is cutting the top leaves off of three strawberries / a man slices the top off a strawberry / a man slicing the greens off of strawberries / a person is cutting the stems off of strawberries / a person is cutting the tops off strawberries with a knife / a person is slicing off the stems of strawberries / a person is slicing strawberries / a woman chopping off the strawberrys stalks / a woman chopping the stalks off of strawberries / a woman is chopping the stem off from prewashed strawberries using a knife / a woman is cutting the stems off strawberries / a woman is preparing strawberry daifuku / a woman is removing head of strawberry / a woman is using a knife to remove the head from strawberries / a woman removing a stems from strawberry / a woman removing the stems of strawberries / cutting the top postion leafy postion strawberry fruit / someone cutting the stems off of strawberries / someone is slicing the tops off some strawberries / the man removed the stems over pre washed strawberries / the person is slicing the top off the strawberries / the top leafy part of the strawberry is being cut / the top part of the strawberry is being removed / the tops of three strawberries are cut off / the women is making strawberry daifuku / woman cutting tops off strawberries / a woman is cutting unwanted part of strawberry fruits / a woman showing how to make strawberry daifuku / cutting the straw berry

Item label	Captions
YI0cxuNcq8_262_272	a woman demonstrating abdominal exercises / a girl is exercising / a girl is exercising / a lady did exercises on a mat / a woman does exercises on a floor mat / a woman exercise her ab muscles on a floor mat / a woman is doing exercise in the room / a woman is doing exercises on the floor / a woman is doing exercises / a woman is exercising her abs / a woman is exercising on the floor / a woman is exercising on the floor / a woman is exercising / a woman is exercising / a woman is exercising / a woman is laying on her back on a mat and she is exercising / a woman is performing some floor exercises / a woman is working out / a woman lying on her back on an exercise mat is moving her legs back and forth and her head from side to side / the girl exercised on a mat / the woman is exercising / a girl doing excercise on the floor / a women doing exercise / the women is excising / the woman is doing cyclic exercise / a woman is doing exercise / the lady is showing 15minute abs workout / a woman is exercising / a woman is demonstrating ab exercises / a lady is doing exercise / a woman is exercising / a girl is practicing the yoga on the floor / a woman is doing healthful exercise / a woman is laying on the floor doing abdominal exercises / a girl is doing exercise / a woman is working out on a yoga pad / a woman is doing exercise on floor / a lady is doing a excercise / a woman is exercising / woman doing exercise / one girl doing exercise / a woman is doing aerobics / a woman doing some sort of abe workout / the lady demonstrates the tips of aerobics / a woman is exercising on a blue yoga mat / a woman is excersizing / a lady is exercising / women showing how to do exercise / a woman playing workout / a young woman is exercising on a mat / a woman is exercising on a mat / a lady is doing workout / she is excise / a woman is doing exercise on a floor mat / a woman exercises

C Small sample from SimVerb-3500

Word1	Word2	Similarity Score
reply	respond	9.79
take	remove	6.81
feed	starve	1.41

Table 2: A sample of word pairs as found in SimVerb-3500 [20], which is one of the benchmarks we use for our study.